XIAOMENG XU

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EDUCATION

Stanford University 09/2023-now
PhD Candidate in Electrical Engineering, Advisor: Shuran Song

Stanford University 09/2023-06/2025
Masters in Electrical Engineering

Tsinghua University 08/2019-06/2023
Bachelor of Engineering in Automation (GPA 3.96/4.0), Bachelor of Arts in Industrial Design

RESEARCH INTEREST

My research explores how machine learning expands the robot design space, by co-designing robot hardware together with the policies that control it. This spans generative models for robot morphology and hardware-software co-design, whole-body vision and distributed sensing, and scalable robot-free data collection for learning from human demonstrations.

PUBLICATIONS

1. Yimeng Qin*, **Xiaomeng Xu***, William Heap, Aditi Oak, Shuran Song, Allison Okamura, *PanoVine: Whole-Body Visuomotor Control for Soft Growing Vine Robot.* [Paper]
2. Dian Wang*, Jisang Park*, **Xiaomeng Xu**, Han Zhang, Shuran Song, Jeannette Bohg, *Mixture of Frames Policy: Multi-Frame Action Denoising for Bimanual Mobile Manipulation.* [Paper]
3. **Xiaomeng Xu**, Jisang Park, Han Zhang, Eric Cousineau, Aditya Bhat, Jose Barreiros, Dian Wang, Jeannette Bohg, Shuran Song, *HoMMI: Learning Whole-Body Mobile Manipulation from Human Demonstrations.* Robotics: Science and Systems (RSS 2026). [Paper]
4. Ryan Punamiya, Simar Kareer, Zeyi Liu, Josh Citron, Ri-Zhao Qiu, Xiongyi Cai, Alexey Gavryushin, Jiaqi Chen, Davide Liconti, Lawrence Y. Zhu, Patcharapong Aphiwetsa, Baoyu Li, Aniketh Cheluva, Pranav Kuppili, Yangcen Liu, Dhruv Patel, Aidan Gao, Hye-Young Chung, Ryan Co, Renee Zbizika, Jeff Liu, **Xiaomeng Xu**, Haoyu Xiong, Geng Chen, Sebastiano Oliani, Chenyu Yang, Xi Wang, James Fort, Richard Newcombe, Josh Gao, Jason Chong, Garrett Matsuda, Aseem Doriwala, Marc Pollefeys, Robert Katzschmann, Xiaolong Wang, Shuran Song, Judy Hoffman, Danfei Xu, *EgoVerse: An Egocentric Human Dataset for Robot Learning from Around the World.* Robotics: Science and Systems (RSS 2026). [Paper]
5. Hojung Choi*, Yifan Hou*, Chuer Pan, Seongheon Hong, Austin Patel, **Xiaomeng Xu**, Mark Cutkosky, Shuran Song, *In-the-Wild Compliant Manipulation with UMI-FT.* IEEE International Conference on Robotics and Automation (ICRA 2026).[Paper]
6. **Xiaomeng Xu***, Yifan Hou*, Chendong Xin, Zeyi Liu, Shuran Song, *Compliant Residual DAgger: Improving Real-World Contact-Rich Manipulation with Human Corrections.* Neural Information Processing Systems (NeurIPS 2025), Best Paper Award at the Human-to-Robot Workshop (CoRL 2025). [Paper]
7. Haoyu Xiong, **Xiaomeng Xu**, Jimmy Wu, Yifan Hou, Jeannette Bohg, Shuran Song, *Vision in Action: Learning Active Perception from Human Demonstrations.* Conference on Robot Learning (CoRL 2025). [Paper]
8. **Xiaomeng Xu**, Dominik Bauer, Shuran Song, *RoboPanoptes: The All-Seeing Robot with Whole-body Dexterity.* Robotics: Science and Systems (RSS 2025). [Paper]

9. **Xiaomeng Xu**, Huy Ha, Shuran Song, *Dynamics-Guided Diffusion Model for Sensor-less Robot Manipulator Design.* Conference on Robot Learning (CoRL 2024), Best Machine Learning Paper Award at the Morphology-Aware Policy and Design Learning Workshop (CoRL 2024). [Paper]

10. **Xiaomeng Xu***, Yanchao Yang*, Kaichun Mo, Boxiao Pan, Li Yi, Leonidas Guibas, *JacobiNeRF: NeRF Shaping with Mutual Information Gradients.* Conference on Computer Vision and Pattern Recognition (CVPR 2023). [Paper]

11. Yun Liu*, **Xiaomeng Xu***, Weihang Chen, Haocheng Yuan, He Wang, Jing Xu, Rui Chen, Li Yi, *Enhancing Generalizable 6D Pose Tracking of an In-Hand Object with Tactile Sensing.* Robotics and Automation Letters (RAL 2023), IEEE International Conference on Robotics and Automation (ICRA 2024). [Paper]

12. Xueyi Liu, **Xiaomeng Xu**, Anyi Rao, Chuang Gan, Li Yi, *AutoGPart: Intermediate Supervision Search for Generalizable 3D Part Segmentation.* Conference on Computer Vision and Pattern Recognition (CVPR 2022). [Paper]

13. Guanhong Liu, Tianyu Yu, Zhihao Yao, Haiqing Xu, Yunyi Zhang, Xuhai Xu, **Xiaomeng Xu**, Mingyue Gao, Qirui Sun, Tingliang Zhang, Haipeng Mi, *ViviPaint: Creating Dynamic Painting with a Thermochromic Toolkit.* Multimodal Technologies and Interaction (MTI 2022). [Paper]

* authors with equal contribution

INTERNSHIP EXPERIENCE

Frontier AI and Robotics Team @ Amazon — 06/2026-09/2026
Research Intern, Mentors: Prof. Guanya Shi, Prof. Carlo Sferrazza — *San Francisco, CA*

· Learning humanoid whole-body loco-manipulation from human demonstrations

Large Behavior Models Team @ Toyota Research Institute — 06/2025-09/2025
Research Intern, Mentor: Eric Cousineau — *Cambridge, MA*

· Scaling whole-body mobile manipulation with egocentric human demonstrations

Stanford Geometric Computing Lab — 06/2022-08/2022
Research Intern, Advisor: Prof. Leonidas Guibas — *Stanford, CA*

· Chinese Undergraduate Visiting Research Program (UGVR).
· Shaping a NeRF to encode mutual correlations of a scene via aligning Jacobians.

AWARDS

NeurIPS 2025 Scholar Award (Travel Grant)	10/2025
Best Paper Award at the Human-to-Robot Workshop (Conference on Robot Learning 2025)	09/2025
Stanford Interdisciplinary Graduate Fellowship (Awarded to outstanding doctoral students engaged in interdisciplinary research at Stanford University)	05/2025-now
Best Machine Learning Paper Award at the Morphology-Aware Policy and Design Learning Workshop (Conference on Robot Learning 2024)	11/2024
CVPR 2023 Travel Grant	06/2023
Outstanding Graduate (Awarded to top 2% graduates at Tsinghua University)	06/2023
Comprehensive Excellence Award (Scholarship awarded by Tsinghua University, top 5%)	10/2022
National Scholarship (Highest scholarship awarded by Chinese Government, top 0.1%)	10/2021
129 Scholarship (Highest scholarship for sophomores at Tsinghua University, top 1%)	10/2020

INVITED TALKS

Robots That See and Act with Their Whole Body @ Graphics Cafe, Stanford 05/2026
From Human Demonstrations to Human Corrections @ RCHI Lab, CMU 04/2026
HoMMI: Learning Whole-Body Mobile Manipulation from Human Demonstrations @ LeCAR Lab, CMU 04/2026
Learning Whole-Body Manipulation via Hardware-Policy Co-design @ SJTU 03/2026
Embodiment-Aware Robot Learning: Co-designing Robot Morphology, Sensing, and Control @ The Helping Hands Lab, Northeastern University 08/2025

SERVICES

Teaching assistant of EE/CS227: Robot Perception (Fall 2024-2025)

Co-organizer of RSS 2025 1st Workshop on Robot Hardware-Aware Intelligence, ICRA 2026 Workshop on Reinforcement Learning in the Era of Imitation Learning, RSS 2026 Diffusion for Robot Learning Workshop

Reviewer of RSS, CoRL, T-RO, RA-L, ICRA, IROS, NeurIPS, ICLR, ICML, CVPR, AAAI

TECHNICAL SKILLS

Computer Languages	Python, C/C++, MATLAB, Verilog/VHDL
Tools	ROS, MuJoCo, SolidWorks, AutoCAD, Rhino, Multisim, Quartus
Hardware	3D printing, mechatronics, mechanical design, woodcraft